

Blind Turbulence Mitigation Based on Ultra-Low-Latency Energy-Efficient Photonic Computing Processor: A Field Trial for Real-Time Underwater Links with 6D-AUV

Geyang Wang^(1, 2, *), Yuxin Wang^(3, *), Jiacheng Ruan⁽⁴⁾, Weipeng Zhang⁽³⁾, Joshua C. Lederman⁽³⁾, Caiming Sun⁽⁴⁾, Bhavin J. Shatri⁽⁵⁾, Lian-Kuan Chen⁽¹⁾, Paul R. Prucnal⁽³⁾

⁽¹⁾ Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong SAR.

⁽²⁾ TÉLÉCOM Paris, Institut Polytechnique de Paris (IP Paris), 91120 Palaiseau, France.

⁽³⁾ Department of Electrical and Computer Engineering, Princeton University, Princeton, NJ 08544, USA

⁽⁴⁾ Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), The Chinese University of Hong Kong, Shenzhen, China,

⁽⁵⁾ Department of Physics, Engineering Physics & Astronomy, Queen's University, Kingston, Canada

*contributed equally to this work, {corresponding author: wg022@ie.cuhk.edu.hk}

Abstract We demonstrate a CSI-free blind turbulence-whitening equalizer (BTWE) using a novel compact modulation-and-weighting microring resonator-based photonic computing processor. Through 6D-AUV underwater field trials, photonic BTWE achieved orders-of-magnitude lower latency and energy consumption compared to advanced DSP with ~50% data-rate enhancement, successfully mitigating turbulence-induced power fluctuations. ©2026 The Author(s)

Introduction

Free-space optical (FSO) communications, encompassing satellite-to-ground links, 6G fronthaul, and underwater wireless optical communications (UWOC), have emerged as pivotal technologies for high-speed, long-range connectivity [1-3]. However, its performance is severely compromised by atmospheric or oceanic turbulence, which induces beam wandering and wavefront distortion, leading to significant optical power fluctuations and fading [4, 5]. To mitigate these effects, various solutions have been explored. On the hardware side, adaptive optics (AO) systems like deformable mirrors can effectively compensate for wavefront distortions but are often cost-prohibitive, bulky, and mechanically complex for deployment in compact terminals [6, 7]. On the software side, electronic digital signal processing (DSP) is widely used. Conventional algorithms frequently require precise channel state information (CSI) or pilot, which occupies valuable bandwidth and faces challenges in fast-varying turbulence scenarios where the channel coherence time is extremely short [8-10]. Moreover, complex electronic DSP incurs excessive computational overhead, resulting in unsustainable energy consumption and high latency [11, 12]. Such bottlenecks impede miniaturization for resource-constrained mobile or underwater platforms [13].

In this paper, we first propose a novel CSI-free Blind Turbulence-Whitening Equalizer (BTWE). The BTWE effectively flattens the signal-to-noise ratio (SNR) distribution in the time domain, significantly mitigating the severe power fluctuations induced by turbulence. To overcome the computational bottleneck of traditional electronic DSP

solutions, we mathematically formulate the BTWE as a matrix-matrix multiplication (MMM), enabling its realization via a compact and scalable photonic computing architecture [14]. To this end, we design a hardware architecture based on modulation-and-weighting microring resonators (MaW-MRRs), which eliminates inter-element spectral alignment and discrete interconnects, drastically reducing footprint. By exploiting the Lorentzian derivative for energy-efficient weight tuning, our chip achieves a remarkable efficiency of 105 TOPs/W, offering a superior power-saving alternative to conventional MRR structures.

Furthermore, we report the first photonic-computing-accelerated blind equalizer for real-time field trial in turbulent UWOC, employing integrated 6-degree-of-freedom (6D) dynamic autonomous underwater vehicle (AUV). By performing weighted summation in the analog optical domain, our photonic computing system achieves orders-of-magnitude performance improvements in latency and energy consumption over conventional electronic DSPs. Experimental results verify that the proposed scheme achieves a ~50% data rate enhancement compared to conventional orthogonal frequency division multiplexing (OFDM) schemes under real-world underwater turbulence with severe power fluctuations.

Principle

For a block of M OFDM symbols with each symbol comprising N subcarriers, Eq. (1) defines the BTWE encoding for the p -th subcarrier of the i -th OFDM symbol:

$$\tilde{x}_{p,q} = \sum_i (x_{p,i} \times c_{i,q}) \quad i, q = 1, \dots, M, \text{ and } p = 1, \dots, N. \quad (1)$$

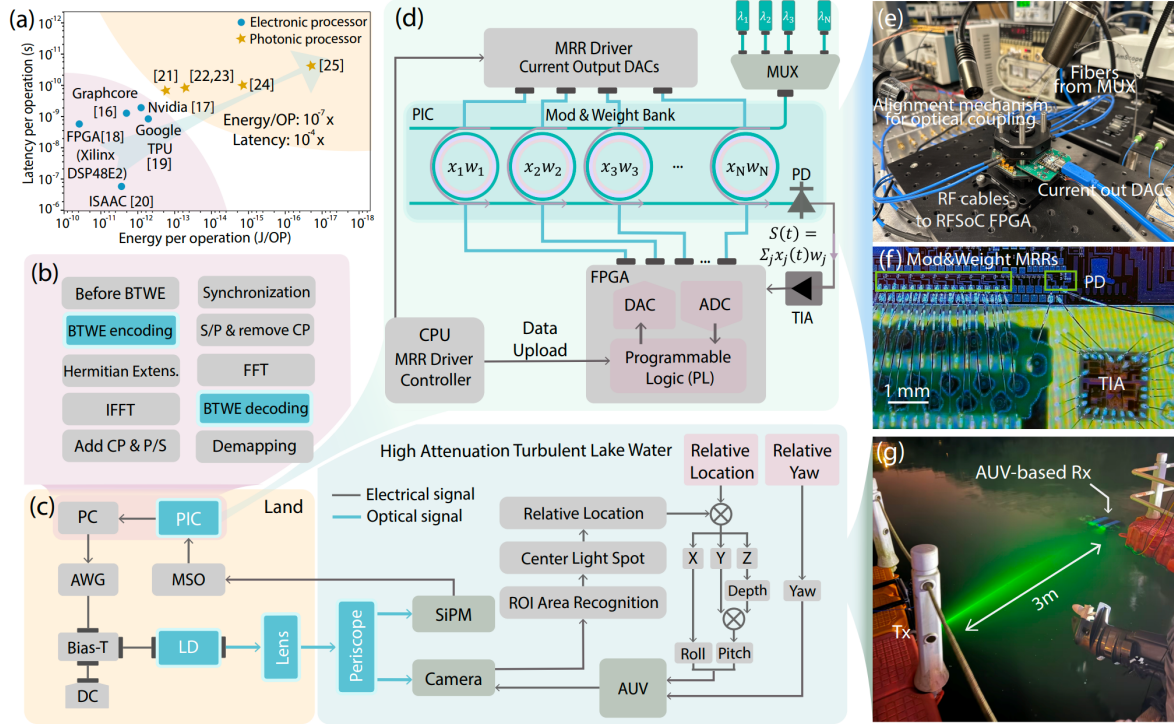


Fig. 1: (a) Illustration of the comparison between the electronic computing and photonic computing processors. (b) The diagram and (c) the experimental setup of dynamic underwater communication system. (d) Schematic diagram of photonic system. (e) Fully packaged photonic integrated circuit (PIC). (f) Micrograph of the PIC. (g) Illustration of field trial. (MRR: Microring Resonator, LD: Laser Diode, PD: Photodetector, SiPM: Silicon Photomultiplier, MUX: Multiplexer, DAC: Digital-to-Analog Converter, ADC: Analog-to-Digital Converter, TIA: Trans-Impedance Amplifier, DC: Direct Current, AWG: Arbitrary Waveform Generator, MSO: Mixed Signal Oscilloscope, PC: Personal Computer, ROI: Region of Interest)

where $x_{p,i}$ is the signal at the i -th OFDM symbol on the p -th subcarrier before the BTWE encoder, and $c_{i,q}$ is phase rotation coefficients. This interleaving-like process utilizes the ideal periodic autocorrelation property of perfect sequences (PS) to reallocate the phase and energy across different time slots [15]. Unlike traditional bit-interleaving, this method spreads cross-symbol information to generate randomized phase profiles and energy levels, effectively flattening the time-domain SNR distribution without requiring CSI. While the CSI-free approach successfully mitigates power fluctuations through whitening, its inherent computational overhead results in excessive latency. By casting this process as a matrix multiplication from Eq. (1) to Eq. (2), it becomes feasible to offload the computation to a PIC.

$$\begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,M} \\ x_{2,1} & x_{2,2} & \dots & x_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \dots & x_{N,M} \end{bmatrix} \times \begin{bmatrix} c_{1,1} & c_{1,2} & \dots & c_{1,M} \\ c_{2,1} & c_{2,2} & \dots & c_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ c_{M,1} & c_{M,2} & \dots & c_{M,M} \end{bmatrix} \quad (2)$$

As shown in Fig. 1(a), photonic processors have been proposed as a promising alternative to DSPs. In Fig. 1(d) to 1(f), we implement a compact photonic computing processor using a bank of 10 MaW-MRRs with 51-GHz channel spacing. By co-integrating PN-junction modulation and thermo-optic tuning within a single device, this

unified architecture eliminates discrete modulators and interconnect overhead while mitigating fabrication-induced wavelength mismatch. Operating in the small-signal regime, the MaW-MRR bank performs linearized analog weighted summation directly in the optical domain. The photonic system functions as an analog precoder, supporting WDM-based parallel processing with 9.7-GHz bandwidth and picosecond-scale latency. This PIC-based platform provides real-time linear transformations in high-speed links. The experimental setup consists of a real-time tracking underwater link between a fixed transmitter and an AUV-mounted receiver, where turbid lake water further exacerbates the turbulence effects [26]. At the transmitter, an AWG produces signals, which modulate a 520-nm laser diode via a bias tee. The data rate ranges from 100 to 175 Mbit/s, with an underwater communication link of 3 meters. BPSK modulation is applied to each OFDM payload subcarrier, and each package consists of 256 OFDM symbols. Fig. 1(b), (c), and 1(g) show the experimental configuration and the field trial site, respectively. The mobile AUV maintains line-of-sight (LOS) using a 6D-based stabilization and tracking system (Fig. 1). An onboard camera monitors the incoming laser spot's position and shape to estimate misalignment. This data is processed in real-time by a controller that

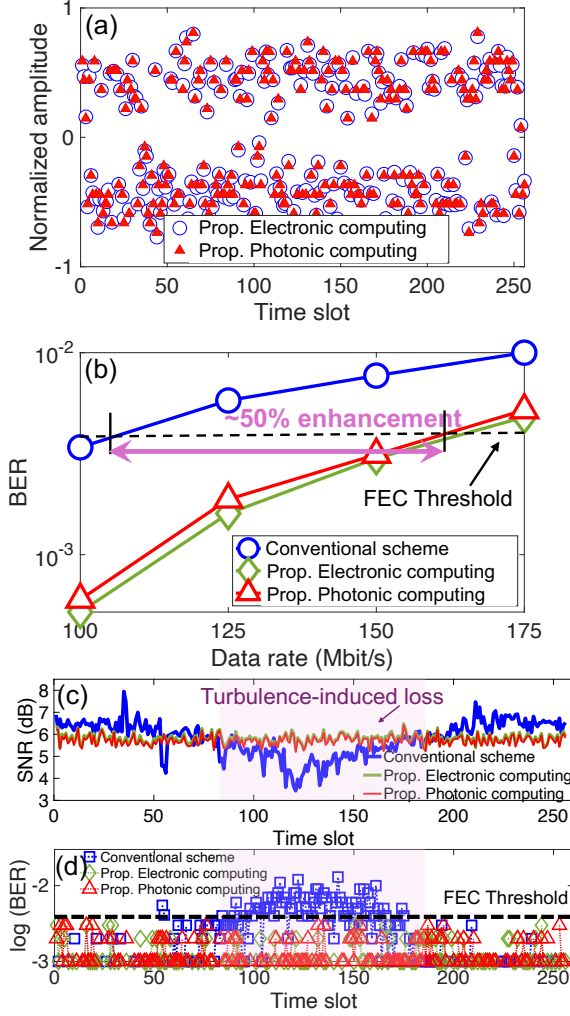


Fig. 2: (a) Experimental results of de-precoder implementation (matrix-matrix multiplication) using photonic computing and electronic computing. (b) BER performance evaluation under real-time tracking with different data rates for various schemes. (c) The SNR and (d) BER versus the time slot under conventional and proposed schemes.

manoeuvres the AUV across all 6D. This setup ensures continuous beam tracking.

Experimental results

Fig. 2(a) demonstrates successful BPSK recovery, with the photonic de-precoder matching the performance of the electronic implementation at reduced hardware overhead. Fig. 2(b) illustrates the BER performance comparison between the proposed photonic computing system and existing benchmarks as the data rate increases. Across all tested data rates, the proposed scheme with real-time tracking exhibits substantial performance gains compared to the conventional scheme. Notably, while the photonic BTWE achieves performance comparable to the electronic one, it does so with significantly reduced energy and latency. At the Forward Error Correction threshold of 3.8×10^{-3} , the proposed scheme enables a $\sim 50\%$ increase in the achievable data rate relative to the conventional real-time tracking

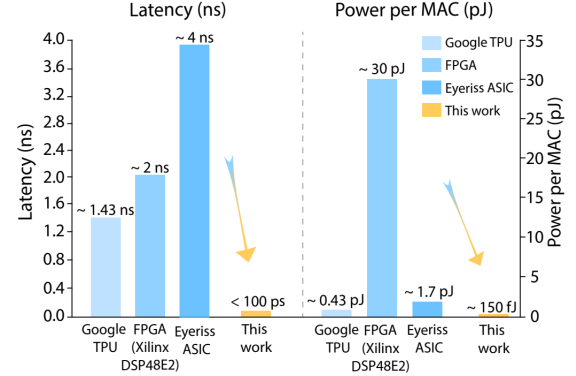


Fig. 3: Latency and power per MAC comparison between advanced electronic DSP and proposed photonic scheme.

scheme. Fig. 2(c) and (d) compare time-domain SNR profiles and BER performance across conventional OFDM and the proposed schemes under power fluctuations. While conventional real-time tracking suffers from localized SNR degradation and instantaneous BER surges, the proposed electronic and photonic schemes maintain a significantly flatter SNR profile, effectively mitigating turbulence-induced burst errors. The photonic implementation achieves nearly identical BER to its electronic counterpart while drastically reducing energy consumption and processing latency. Fig. 3 illustrates the comparison of latency and energy efficiency between electronic and photonic processors. The platform's performance is characterized by an energy efficiency of approximately 150 femtojoules (fJ) per multiply-accumulate (MAC) operation and a tens-of-picosecond processing latency. It is evident that the proposed photonic computing processor achieves orders-of-magnitude improvements in both processing latency and energy efficiency compared to various state-of-the-art electronic benchmarks.

Conclusions

In this work, we have successfully demonstrated the first photonic-enhanced blind equalizer, validated through a field trial for real-time underwater links involving a 6D-AUV, that addresses the critical challenges of turbulence-induced power fluctuations in high-speed UWOC links. By implementing a novel CSI-free BTWE algorithm on a high-performance MaW-MRR photonic processor, we effectively overcame the electronic bottleneck inherent in conventional DSP. Our unified PIC design achieved a significant advancement in performance, reducing processing latency to tens of picoseconds and energy consumption to ~ 150 femtojoules per operation. The first-of-its-kind photonic blind equalizer also realizes $\sim 50\%$ data rate improvement under adverse real-world underwater environments. This approach provides a scalable and energy-efficient pathway for real-time signal processing in next-generation mobile optical wireless networks, ground-to-satellite FSO links, and 6G fronthaul.

Acknowledgements

This research is supported by HKSAR RGC (GRF-14201324 and C5078-24GF), the National Science Foundation (NSF) (ECCS-2128616 and ECCS-1642962 to P.R.P.), the Office of Naval Research (ONR) (N00014-18-1-2297 and N00014-20-1-2664 P.R.P.), the Defense Advanced Research Projects Agency (HR00111990049 to P.R.P.), the National Natural Science Foundation of China under Grant 62175120 and in part by the Shenzhen Science and Technology Program under Grant JCYJ20220818103011023.

References

- [1] V. V. Vliet, M. Van Den Hout, K. Gümüş, E. Tangdi-ongga and C. Okonkwo, "Experimental Investigation of Availability in a 4.6 Km Terrestrial Urban Coherent Free-Space Optical Communications Link," 2025 European Conference on Optical Communications (ECOC), Copenhagen, Denmark, 2025, pp. 1-4, doi: 10.1109/ECOC66593.2025.
- [2] Shuang Li, Ping Wang, Li Yang, Wenwen Chen, Hanxiao Li, Qian Hu, and A. S. Madhukumar, "Performance evaluation of the vertical UWOC system under jamming over the unified multi-layer turbulence channel," *Opt. Express* 33, 45807-45827 (2025)
- [3] Mehul Malik, Malcolm O'Sullivan, Brandon Rodenburg, Mohammad Mirhosseini, Jonathan Leach, Martin P. J. Lavery, Miles J. Padgett, and Robert W. Boyd, "Influence of atmospheric turbulence on optical communications using orbital angular momentum for encoding," *Opt. Express* 20, 13195-13200 (2012)
- [4] M. A. Kashani, M. Uysal, and M. Kavehrad, "A novel statistical channel model for turbulence-induced fading in free-space optical systems," *Journal of Lightwave Technology*, vol. 33, no. 11, pp. 2303-2312, Jun. 2015. DOI: 10.1109/JLT.2015.2410695
- [5] YANG Jinglin, WANG Yujue, DENG Deying, et al., "Underwater wireless optical communication in turbulent channels: a review," *Infrared and Laser Engineering*, vol. 55, no. 2, p. 20250522, 2026.
- [6] Dhwanil Patel, Momen Diab, Ross Cheriton, Jacob Taylor, Libertad Rojas, Martin Vachon, Dan-Xia Xu, Jens H. Schmid, Pavel Cheben, Siegfried Janz, and Suresh Sivanandam, "End-to-end simulations of photonic phase correctors for adaptive optics systems," *Opt. Express* 32, 27459-27472 (2024)
- [7] Schiavon, M.; Vanzo, A.; Campaci, K.; Maru-landa Acosta, V.; Bonora, S. Multi-Actuator Lens Systems for Turbulence Correction in Free-Space Optical Communications. *Photonics* 2025, 12, 870. <https://doi.org/10.3390/photonics12090870>
- [8] Zhang, R., Hu, N., Zhou, H. et al. Turbulence-resilient pilot-assisted self-coherent free-space optical communications using automatic optoelectronic mixing of many modes. *Nat. Photon.* 15, 743-750 (2021).
- [9] Yan Zhang, Ping Wang, Lixin Guo, Wei Wang, and Hongxin Tian, "Performance analysis of an OAM multiplexing-based MIMO FSO system over atmospheric turbulence using space-time coding with channel estimation," *Opt. Express* 25, 19995-20011 (2017)
- [10] Yan Zhang, Ping Wang, Tao Liu, Lixin Guo, Yutao Li, and Wei Wang, "Performance analysis of a LDPC coded OAM-based UCA FSO system exploring linear equalization with channel estimation over atmospheric turbulence," *Opt. Express* 26, 22182-22196 (2018)
- [11] Fontaine, N. K. et al. Digital turbulence compensation of free space optical link with multimode optical amplifier. In *Proc. European Conference on Optical Communication PD.1.1* (IET, 2019).
- [12] Z.-Y. Zhu, M. Janasik, A. Fyffe, et al., "Compensation-free high-dimensional free-space optical communication using turbulence-resilient vector beams," *Nature Communications*, vol. 12, article no. 1666, 2021.
- [13] Y. Shao, R. Deng, J. He, K. Wu and L. -K. Chen, "Real-Time 2.2-Gb/s Water-Air OFDM-OWC System With Low-Complexity Transmitter-Side DSP," in *Journal of Lightwave Technology*, vol. 38, no. 20, pp. 5668-5675, 15 Oct.15, 2020, doi: 10.1109/JLT.2020.3001864.
- [14] Zhang, W., Wang, Y., Lederman, J.C., Shastri, B.J., and Prucnal, P.R., 2026. Compact, reconfigurable, and scalable photonic neurons by modulation-and-weighting microring resonators. *Elight*, 6(1), p.6.
- [15] G. Wang, D. Zhang, J. Zhao and L. -K. Chen, "A unified precoding-based FTN scheme for communication channels with limited bandwidth," 49th European Conference on Optical Communications (ECOC 2023), Hybrid Conference, Glasgow, UK, 2023, pp. 1095-1098, doi: 10.1049/icp.2023.2445.
- [16] S. Knowles, "Scalable silicon compute," in *Proc. NIPS Workshop Deep Learn. Supercomputer Scale*, Dec. 2017.
- [17] Smith, Ryan. "Nvidia volta unveiled: Gv100 GPU and tesla v100 accelerator announced." *AnandTech*. Accessed: May 10 (2017).
- [18] AMD Xilinx, "UltraScale Architecture DSP Slice User Guide (UG579)," 2023. [Online]. Available: <https://docs.xilinx.com/r/en-US/ug579-ultrascale-dsp>
- [19] Jouppi, N. P. et al. In-datacenter performance analysis of a tensor processing unit. In *Proceedings of the 44th annual international symposium on computer architecture*. 1-12 (Toronto, ON, Canada: ACM, 2017)
- [20] Shafiee, A. et al. ISAAC: A convolutional neural network accelerator with insitu analog arithmetic in crossbars. *ACM SIGARCH Computer Architecture N.* 44, 14-26 (2016).
- [21] Z. Chen, A. Sludds, R. Davis III, I. Christen, L. Bernstein, L. Ateshian, T. Heuser, N. Heermeier, J. A. Lott, S. Reitzenstein, et al., *Nat. Photonics* 2023, 17, 723
- [22] Shen, Y. C. et al. Deep learning with coherent nanophotonic circuits. *Nat. Photonics* 11, 441-446 (2017).
- [23] Tait, A. N. et al. Silicon photonic modulator neuron. *Phys. Rev. Appl.* 11, 064043 (2019).
- [24] Tait, A. N. Silicon photonic neural networks. PhD thesis, Princeton University, Princeton, 2018.
- [25] Nahmias, M. A. et al. Photonic multiply-accumulate operations for neural networks. *IEEE J. Sel. Top. Quantum Electron.* 26, 7701518 (2020).
- [26] Y. Weng, Y. Guo, O. Alkhazragi, T. K. Ng, J. -H. Guo and B. S. Ooi, "Impact of Turbulent-Flow-Induced Scintillation on Deep-Ocean Wireless Optical Communication," in *Journal of Lightwave Technology*, vol. 37, no. 19, pp. 5083-5090, 1 Oct.1, 2019, doi: 10.1109/JLT.2019.2928465.